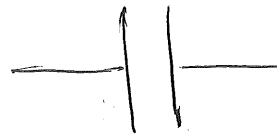


Chapter 6

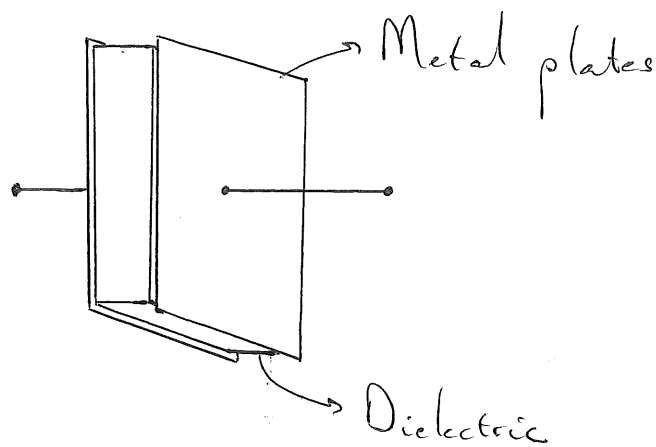
Lecture 1.

①

* Capacitors



- Passive element (see chap 1)
- Stores energy in electric field.
- Capacitor is two conducting plates separated by an insulator (dielectric)



$A \Rightarrow$ surface area of metal plates $[m^2]$

$\epsilon \Rightarrow$ permittivity of the dielectric $[F/m]$

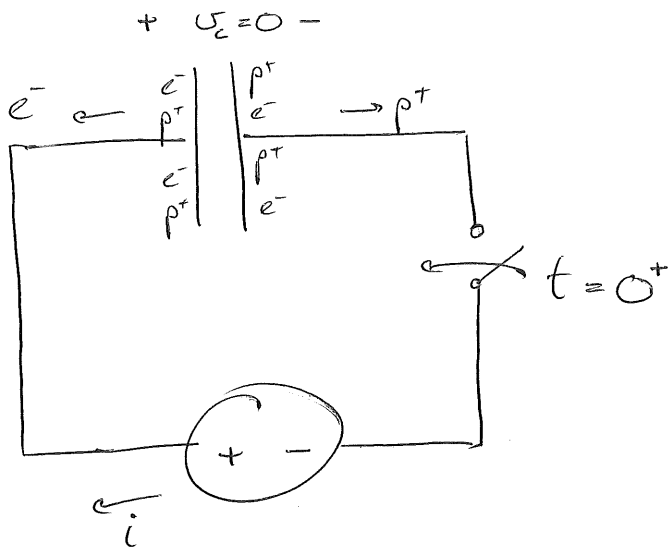
$d \Rightarrow$ distance between plates $[m]$

$$* C = \frac{\epsilon A}{d} \quad [F] = \text{Farad.}$$

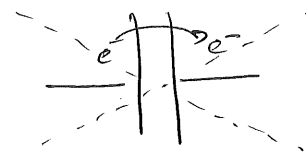
→ What is capacitance?

(2)

* Ratio of charge q on one plate of capacitor and voltage difference between plates. Unit: Farad.

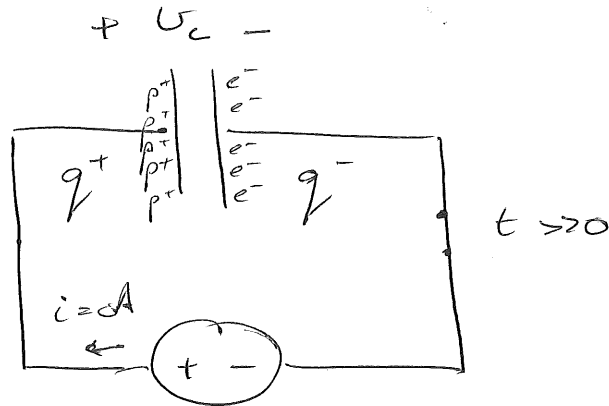


* Note: charge cannot go from one plate to the other!

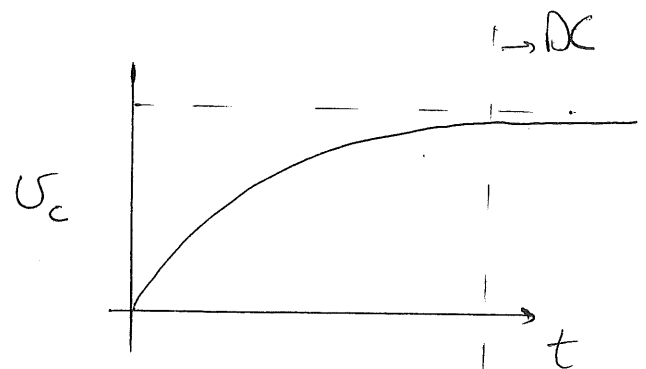
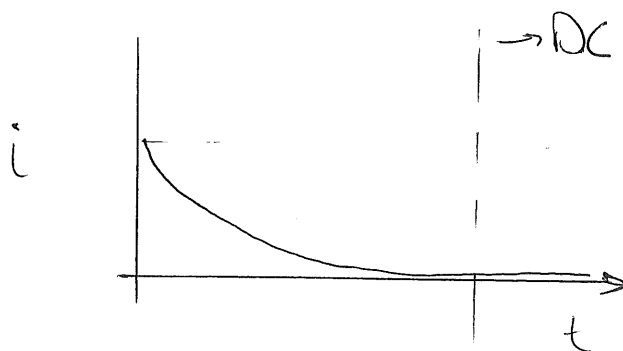


- * At $t < 0$; equal charge on both plates. ($V_C = 0$ V.)
- * At $t = 0^+$, switch is closed. Pathway for charge to flow. Neg charge goes to pos terminal. Pos charge goes to neg terminal of voltage source.
- * Initially there is much charge that can flow. As time goes by, less charge can be drawn from capacitor plates by voltage source.

Therefore at $t \gg 0$, both plates cannot supply neg or pos charge to flow in circuit.

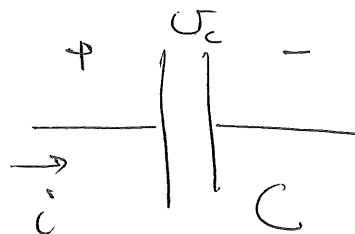


- * Difference in charge between plates means a voltage difference. $V_c \propto q$
- * Voltage across capacitor is directly equivalent to charge on plates.
- * The graphs below show the current and voltage in the circuit as time goes by.



DC-state or Steady-state

(4)

* Equations:

$$\rightarrow C = q/v_c \quad (1)$$

$$\rightarrow i = \frac{dq}{dt} = C \frac{dv_c}{dt} \quad (2) \text{ (From (1))}$$

$$\rightarrow \therefore v_c = \frac{1}{C} \int_{t_0}^t i dt + v(t_0) \quad (3)$$

Voltage across capacitor is the integration of the current through the capacitor from time " t_0 " to a final time " t ".

* Energy and power.

$$\rightarrow p = vi = v \times C \frac{dv}{dt}$$

$$\rightarrow w = \frac{1}{2} C v^2$$

* DC conditions. ($t \gg 0$) (V and I constant)

$$\rightarrow \text{If } v \text{ constant: } i = C \frac{dv}{dt} = 0 \text{ A.}$$

$\therefore$ At DC conditions, capacitor is an open circuit.

Examples:

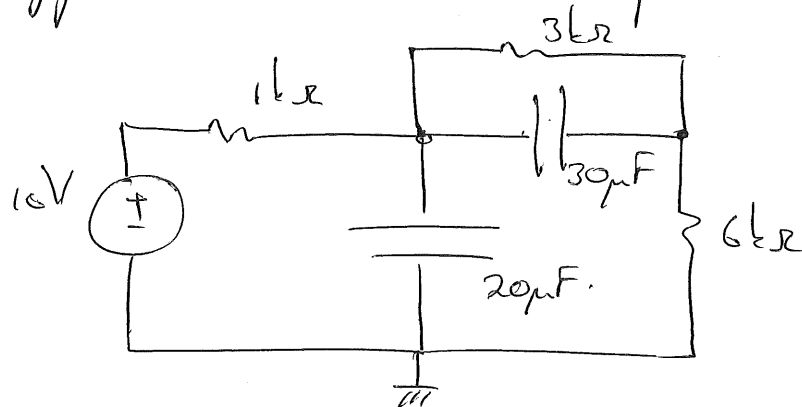
(5)

① $C = 10 \mu\text{F}$ $v_c(t) = 50 \sin(2000t) \text{ V}$
 $i(t) = ?$

② $C = 3 \mu\text{F}$ $q = 0.12 \text{ mC}$
 $V_c = ?$ $\omega_c = ?$

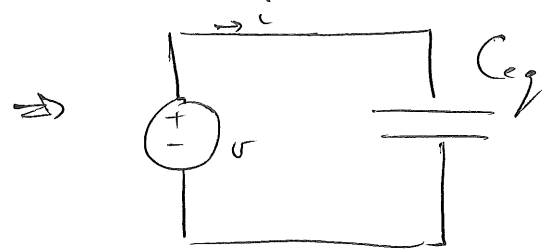
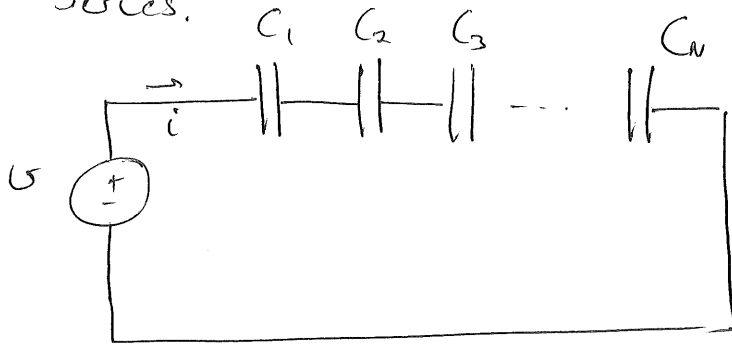
③ $C = 100 \mu\text{F}$ $i(t) = 50 \sin(120\pi t) \text{ mA}$
 $v(1 \text{ ms}) = ?$ $v(5 \text{ ms}) = ?$
Assume $v(0) = 0 \text{ V}$.

④ DC conditions. Calculate ω_1 and ω_2 .
(Energy stored in both capacitors.)



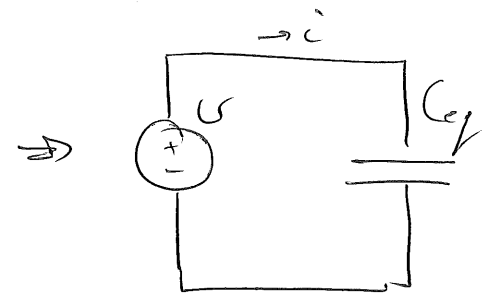
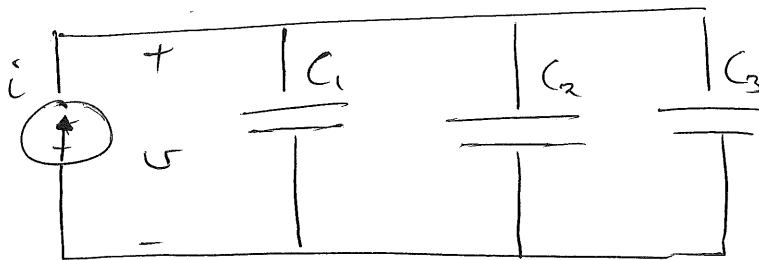
* Series and Parallel capacitors.

→ Series.



$$\frac{1}{C_{eq}} = \frac{1}{C_1} + \frac{1}{C_2} + \frac{1}{C_3} \dots + \frac{1}{C_N}$$

→ Parallel.



$$C_{eq} = C_1 + C_2 + C_3 \dots + C_N$$

Example

$C_{eq} = ?$

